SEMITEC FOUNDATION · [Course name]

Lab 1: [Lab title — what the student builds]

Lab at a glance

Duration	Difficulty	Work in	Version
[e.g. 2 hours]	**Beginner**	**[e.g. individually / pairs]**	**1.0**

Equipment	[board, cables, instruments]
Software	[tools and versions]
Prerequisites	[what the student must know before — link trainings / earlier labs]
Author	[Name, academic title]
Department	[Department]

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1. Overview

[Two or three sentences: what you build in this lab and where it is used in practice.]

Objectives

After this lab you can:

- **[objective 1 — a verb that can be checked]**
- **[objective 2]**
- **[objective 3]**

2. Background

[The theory the student needs: definitions, a diagram, a short example.]

3. Preparation

- **[Tool and version to open; board files to install.]**
- Download the lab files: in SemiTec Academy open this lab → **Files** → *Download all*.
- **[Reading or training to do before the lab.]**

Safety

[Handling rules for the equipment (ESD, power, cables). Remove this box if not needed.]

4. Procedure

Step 1 Set up the project

[What to create / open, which files to add.]

Step 2 Write the design

Files: `src/README.txt`

[What to write. Show the interface or a skeleton:]

Step 3 Simulate

[How to run the testbench and what to look at.]

Expected result: [what the waveform / log must show]

Step 4 Test on the board

[Program the board and test.]

✓ Checkpoint

[What the teacher checks before the student continues.]

5. Results

Table 1: [Caption]

Quantity	Expected	Measured	Unit
[quantity]	[expected value]	_____	[unit]

6. Questions

Q1. [Why …?]

Q2. [What happens if …?]

7. Report and assessment

Hand in:

- [source files]
- [screenshot / photo of the result]
- answers to the questions

Criterion	Weight
[Design works on the board]	40 %
[Testbench and simulation]	30 %
[Answers to the questions]	30 %

8. References and further reading

[1] *[Tool or board user guide]*. [Document number, version]. [Vendor]. [Year].

[2] [Author]. *[Textbook title]*. [Publisher], [Year].